

DTU



Measuring social acceptance and preferences – The case of Carbon Capture and Storage

Overview

- What is social acceptance
- Why do they matter
- How do we measure it
- What are the strengths and the weaknesses
- Empirical results

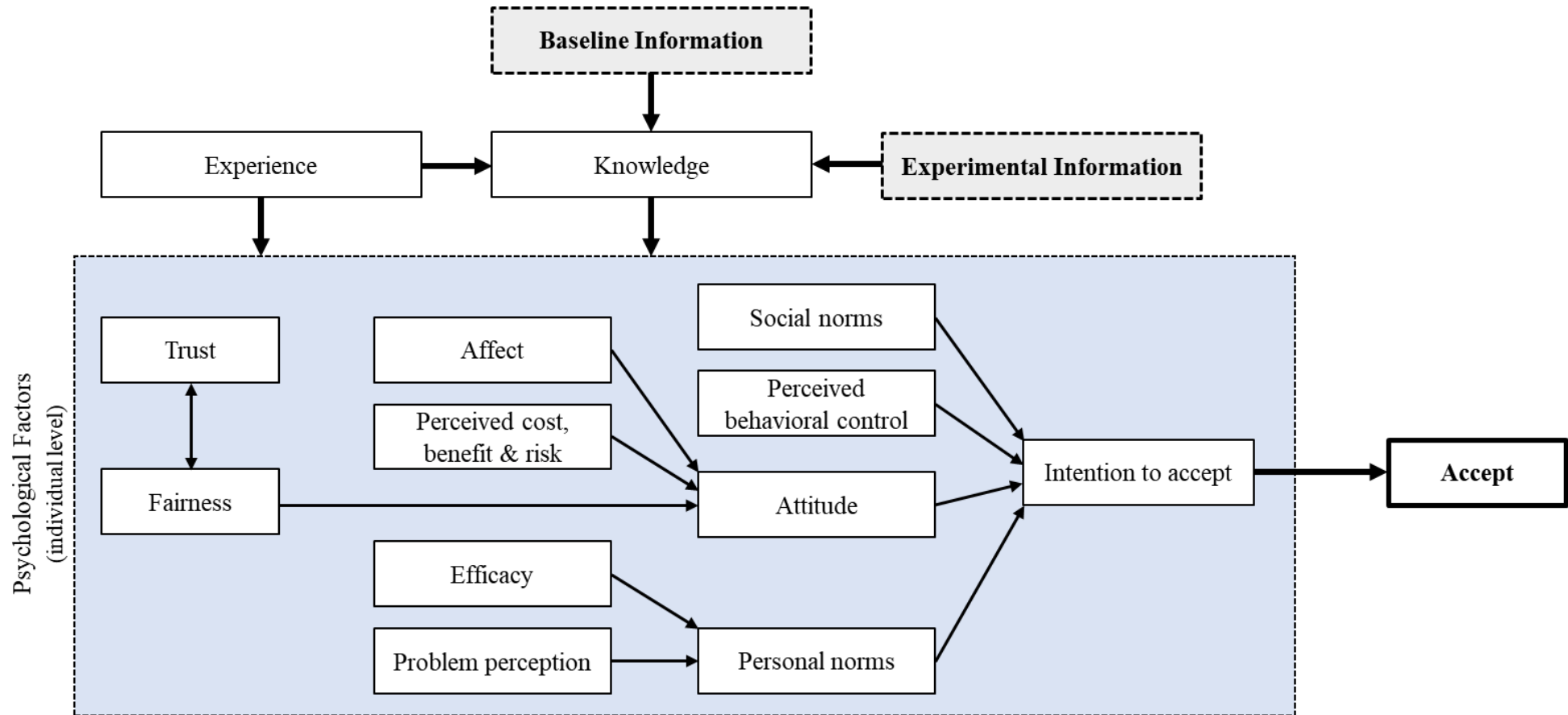
Introduction

- The concept of social acceptance plays a pivotal role in the successful implementation of Carbon Capture and Storage (CCS) initiatives. (Ashworth et al., 2012)
- Social acceptance refers to the willingness of individuals, communities, and stakeholders to support and integrate CCS technologies into society. (Gaede and Rowlands, 2018)
- Social acceptance encompasses public attitudes, perceptions, and beliefs towards CCS projects.
- It involves the integration of technological, economic, environmental, and ethical considerations.
- Understanding social acceptance is crucial for building a sustainable and inclusive transition to a low-carbon future.

Examples of Social Acceptance Questions

- "Do you believe that CCS can effectively reduce carbon emissions and mitigate climate change?"
- "Are you willing to support CCS initiatives despite potential safety concerns?"
- "To what extent do you feel involved in the decision-making process of CCS projects in your community?"
- "Are you concerned about the long-term impacts of CO₂ storage through CCS?"
- "Do you perceive CCS as a practical solution for transitioning to a low-carbon energy future?"

A theoretical model on acceptance based on Huijts et al. (2012) and Zuch and Ladenburg (2023)



Experience, knowledge and information effects

- **Experience:** Direct encounters with CCS projects or related technologies impact perceptions and attitudes.
- **Knowledge:** Understanding of CCS processes, benefits, and risks influences acceptance.
- Generally low level of knowledge
- Baseline information is often provided giving a general introduction to the technology
- *"CO₂ capture and storage (CCS) is a technology where CO₂ is captured at power plants or industries. The CO₂ is transported via pipes/by ship to the storage site, and pumped deep into the subsoil, where nature itself ensures that the CO₂ stays there. In Denmark, we are already testing the possibilities of capturing CO₂ and storing it in two large old oil/gas fields in the North Sea". (Ladenburg et al. 2023)*
- Experimental information
 - Within subject
 - Random Trial Control (RTC) between subjects (causal effects can be estimated)

Psychological Factors

- **Trust:** Confidence in regulatory bodies, industry, and technology developers fosters acceptance.
- **Fairness:** Perceived equitable distribution of benefits and burdens enhances support.
- **Affect:** Emotional reactions and positive associations influence overall acceptance.
- **Perceived Costs, Benefits, and Risks:** Assessment of economic, environmental, and social impacts informs acceptance.
- **Efficacy:** Belief in the effectiveness of CCS contributes to willingness to support.
- **Social and Individual Norms:** Alignment with societal values and personal beliefs shapes acceptance.
- **Attitude:** Favorable opinions driven by perceived benefits, values, and overall assessment.
- **Perceived Behavioral Control:** Belief in one's ability to influence or participate in CCS decisions impacts acceptance.

Understanding the interplay

- These determinants interact and influence each other, forming a complex web of acceptance dynamics.
- Effective communication strategies should address these factors to enhance social acceptance.

Drivers of Social Acceptance

- Environmental Concerns: Acknowledging the urgent need for climate change mitigation and reduced greenhouse gas emissions (Itaoka et al., 2009)
- Economic Opportunities: Recognizing the potential for job creation, industry growth, and economic development. (von Rothkirch and Ejderyan, 2021)
- Technological Innovation: Embracing CCS as a bridge to a cleaner energy future while renewable technologies advance. (Braun et al. 2018)

Challenges to Social Acceptance

- Risk Perception: Addressing fears related to potential accidents, leaks, and long-term consequences. (Karimi & Toika 2014)
- Equity and Justice: Ensuring fair distribution of benefits and addressing potential negative impacts on vulnerable communities. (Karytsas et al. 2023)
- Policy and Regulation: Developing comprehensive frameworks to ensure safety, accountability, and effective governance.

Importance of Social Acceptance

- Successful Implementation: Social acceptance fosters collaboration, support, and buy-in from various stakeholders.
- Reduced Resistance: Overcoming opposition and skepticism through informed public engagement.

Social acceptance on different levels

- **Site or Project Specific:** Tailoring questions to the local context addresses unique community concerns and needs about a specific existing or proposed project. (Oltra et al. 2022)
 - Acceptance of local gas storage facility
 - Acceptance of gas storage transport in local area
- **General Population Questions:** Broader inquiries provide insights into wider societal perceptions and trends towards CCS. Ladenburg et al. (2023)
 - Acceptance of the use of CCS in a broader specified geographical unit (municipality, region, country or EU/US) but not referring to a specific project.

Data sources

- **Surveys:** Structured questionnaires measure public attitudes, concerns, and knowledge about CCS. (Dillman et al., 2014, Karytsas et al. 2023, Ladenburg et al. 2023)
 - Local residence
 - National
 - Survey company respondents
 - Central register respondents

- **Personal Interviews or focus group interviews:** In-depth (group dynamic) conversations allow for nuanced exploration of beliefs and opinions. (Ha-Duong et al., 2009, Newcomer et al., 2015, Saunders et al. 2009, Shackley et al., 2004)
 - Open questions
 - Semi structured questionnaire

Data analysis

- Ordinary least square/order logit regression of result (Greene, 2012)
- Mean comparisons
- Transcription and coding of interviews (Matthew B.Miles et al., 2019; Patton, 2014)
- Topics coding

Investigating Heterogeneity in Acceptance

- Analyzing acceptance variation across socio-economic groups provides valuable insights.
- Factors like gender, age, education, income, and area of residence influence perceptions of CCS
- Examples of Investigative Questions
 - "How does gender impact the perception of CCS benefits and risks?"
 - "Do different age groups have varying levels of support for CCS initiatives?"
 - "Is there a correlation between education level and understanding of CCS technology?"
 - "Does income level influence willingness to invest in CCS projects?"
 - "How does geographic location affect attitudes towards CCS in urban and rural areas?"

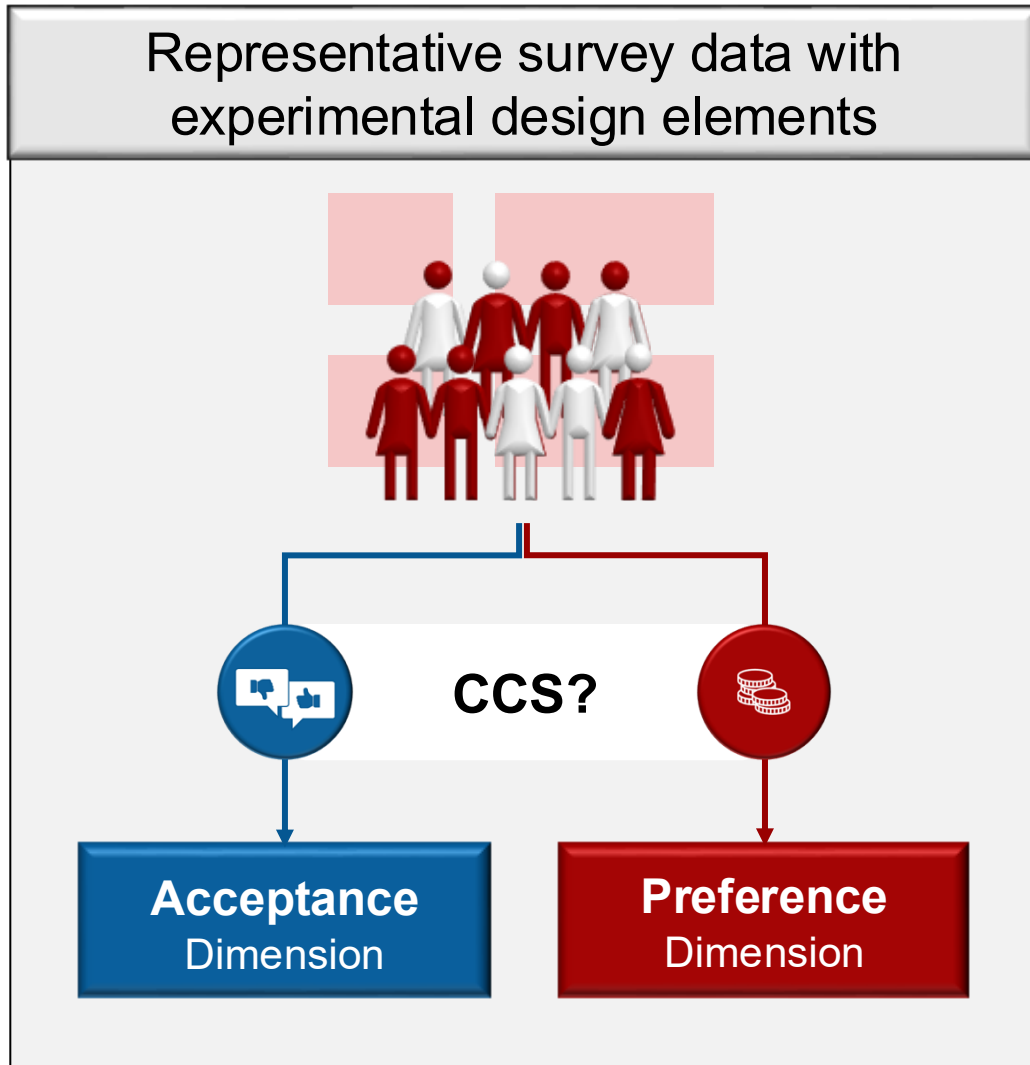
Public Acceptance of various low-carbon energy technologies in Denmark

Research Questions



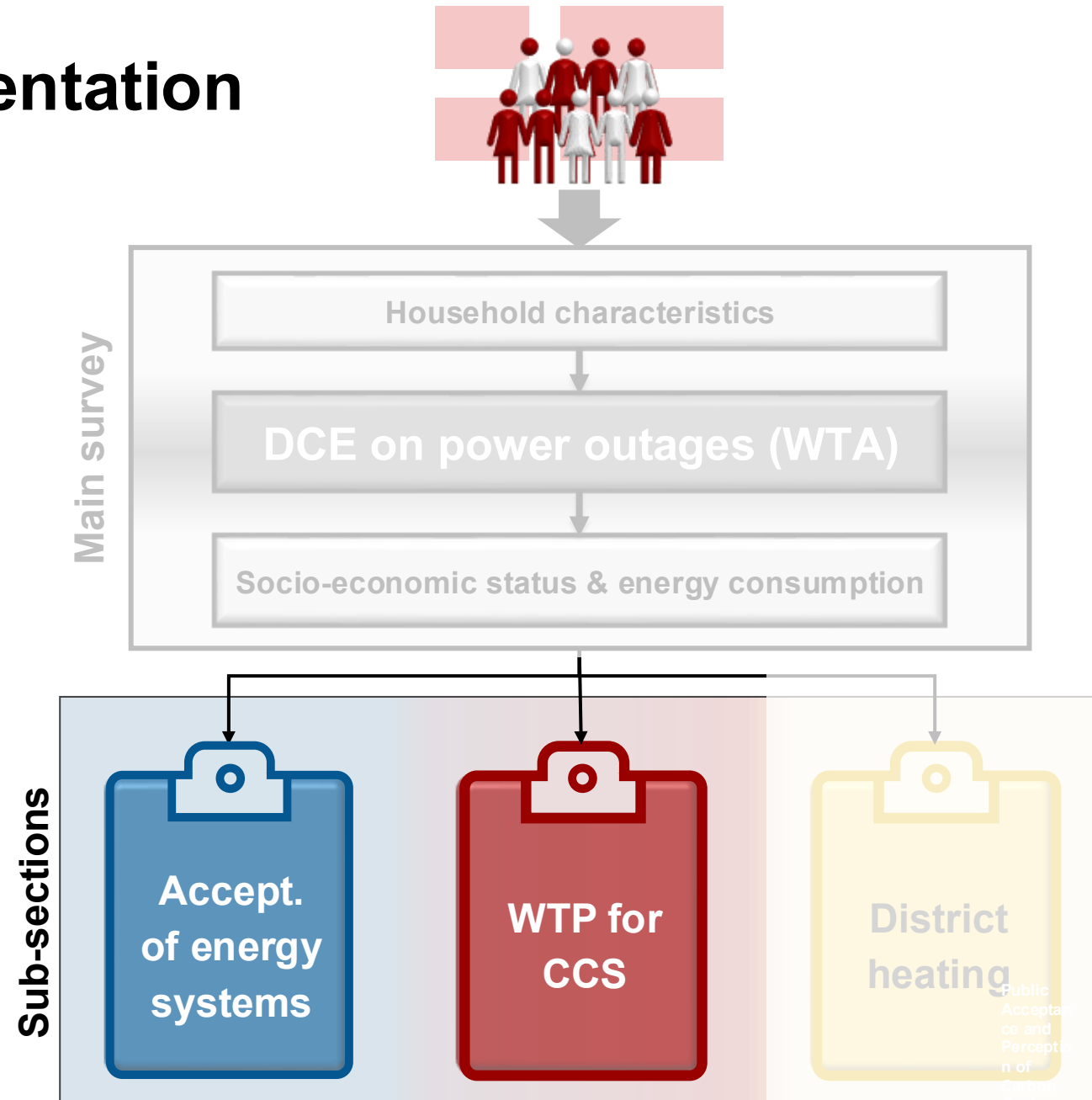
“How and to what extent does information provision affect public responses to CCS?”

Materials & Methods



Survey Design & Implementation

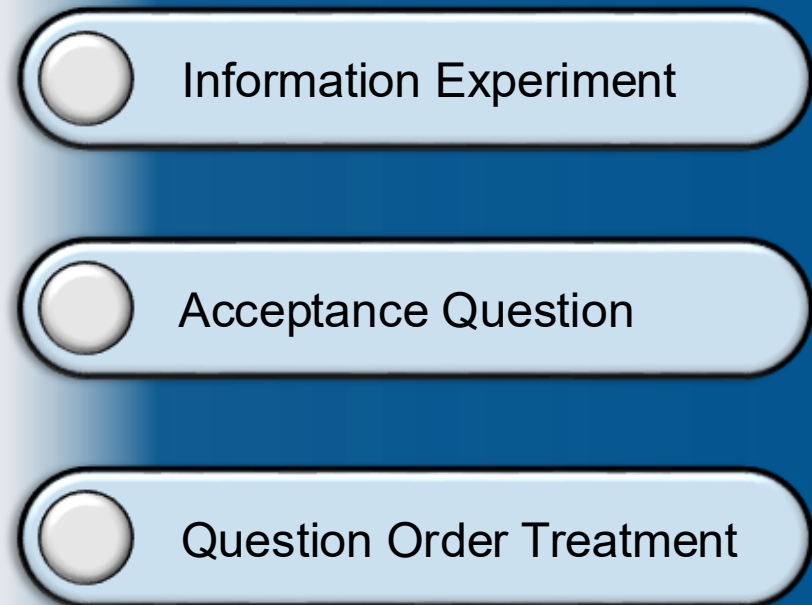
- Large-scale online survey on national energy perception
 - Sent out in August 2022
 - Random sample of 55,000 Danish households
 - Invitation via 'e-boks'
- Total Sample: **12,251** respondents
- Response Rate: **22.3%**



Survey Sub-Section

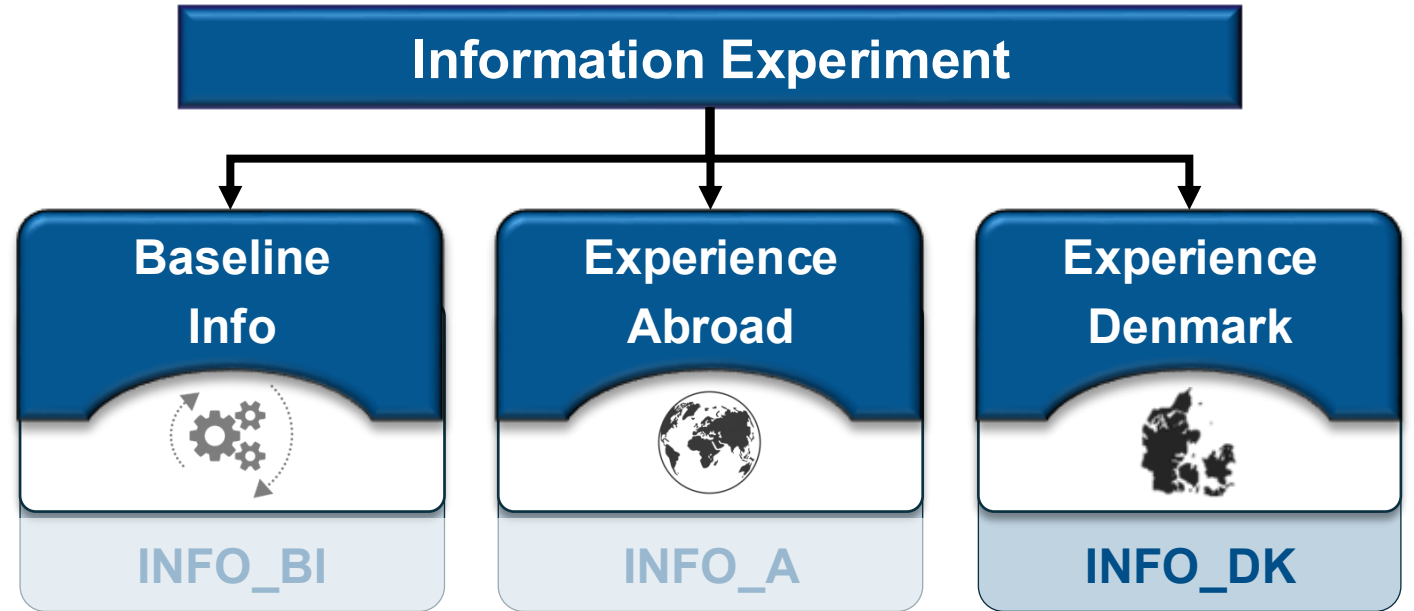
Acceptance of various emission reduction measures

- Randomly assigned to 1/3 of survey respondents
 - Total sample: **4,126 respondents**
- Corrected for invalid acceptance responses:
 - Effective sample: **3,969 respondents**
 - Effective response rate: **21.5%**



Acceptance Survey

- ☒ Information Experiment
- ☐ Acceptance Question
- ☐ Question Order Treatment



Acceptance Survey



Information Experiment



Acceptance Question



Question Order Treatment

Acceptance Question

“To what extent should Denmark use the following measures to reduce CO₂ emissions?”

Very small extent	Small extent	Moderate extent	Large extent	Very large extent
☐	☐	☐	☐	☐

Acceptance Survey

☐ Information Experiment

☒ Acceptance Question

☐ Question Order Treatment

Acceptance Question

“To what extent should Denmark use the following measures to reduce CO₂ emissions?”

Energy Saving Technologies



Carbon Capture & Storage (CCS)



Offshore

Nearshore

Onshore Rural

Onshore Urban

CO₂ Taxes



Acceptance Survey

Information Experiment

Acceptance Question

Question Order Treatment

Question Order Treatment

N → T

T → N

Nuclear Power



District Heating



Wind Power



Offshore

Nearshore

Onshore

Solar Power



Residential Roof

Industrial Buildings

Agricultural Land

Energy Saving Technologies



Carbon Capture & Storage (CCS)



Offshore

Nearshore

Onshore Rural

Onshore Urban

CO₂ Taxes

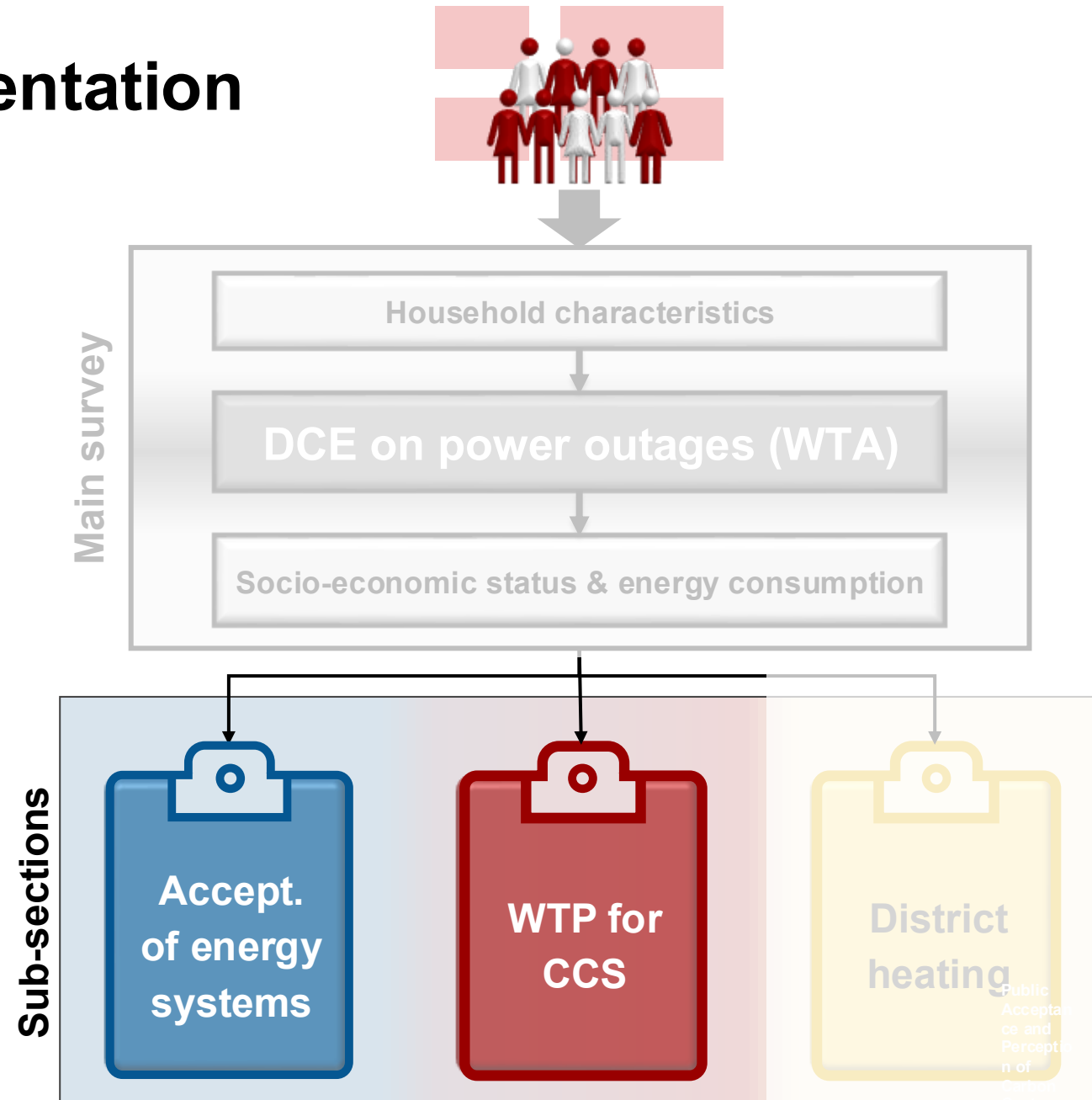


Consumer CO₂ Tax

Producer CO₂ Tax

Survey Design & Implementation

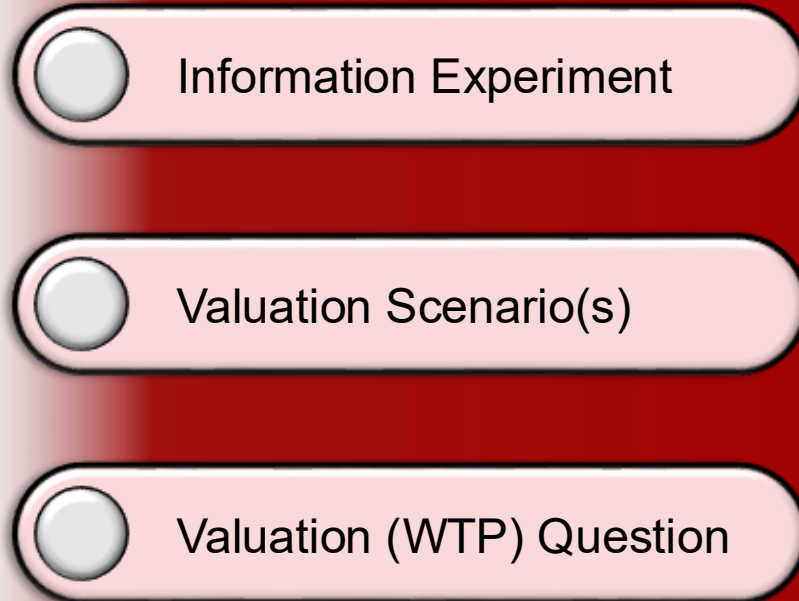
- Large-scale online survey on national energy perception
 - Sent out in August 2022
 - Random sample of 55,000 Danish households
 - Invitation via 'e-boks'
- Total Sample: **12,251** respondents
- Response Rate: **22.3%**



Survey Sub-Section

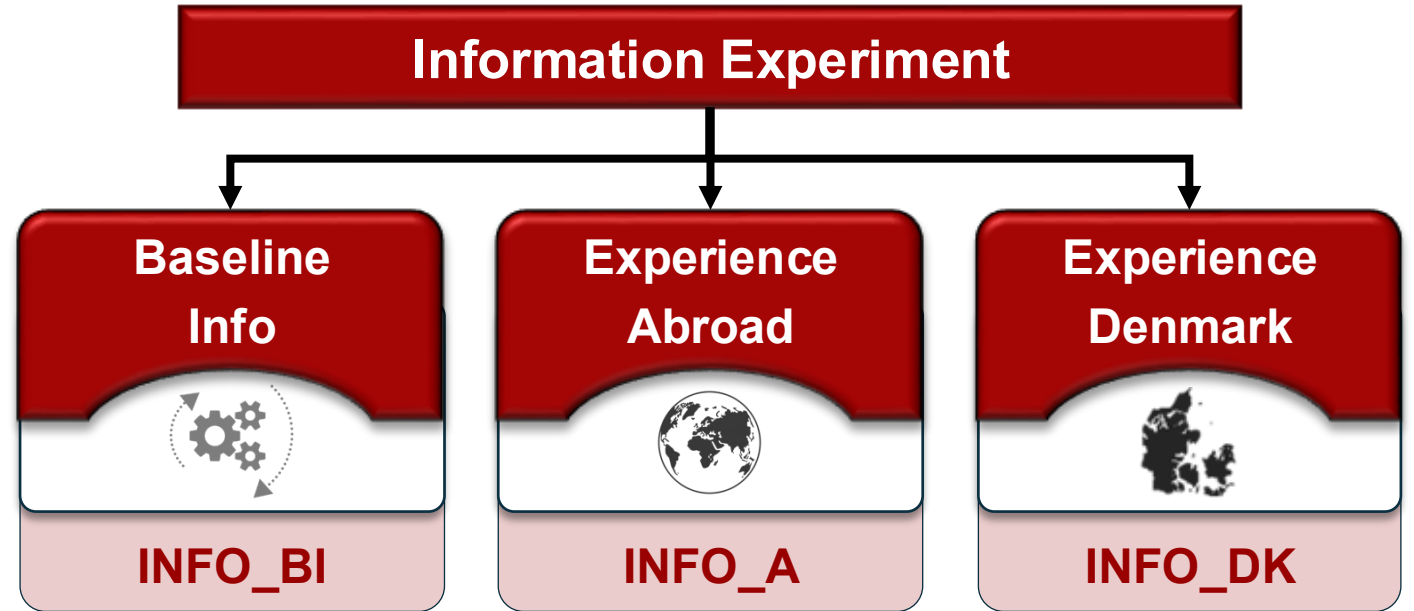
Willingness to Pay (WTP) for CCS

- Randomly assigned to 1/3 of survey respondents
 - Total sample: **4,191 respondents**
- Corrected for protest zero (n=350) and strategic bidders (n=451):
 - Effective sample: **3,390 respondents**
 - Effective response rate: **18.5%**



WTP for CCS Survey

- ☒ Information Experiment
- ☐ Valuation Scenario(s)
- ☐ Valuation (WTP) Question



WTP for CCS Survey

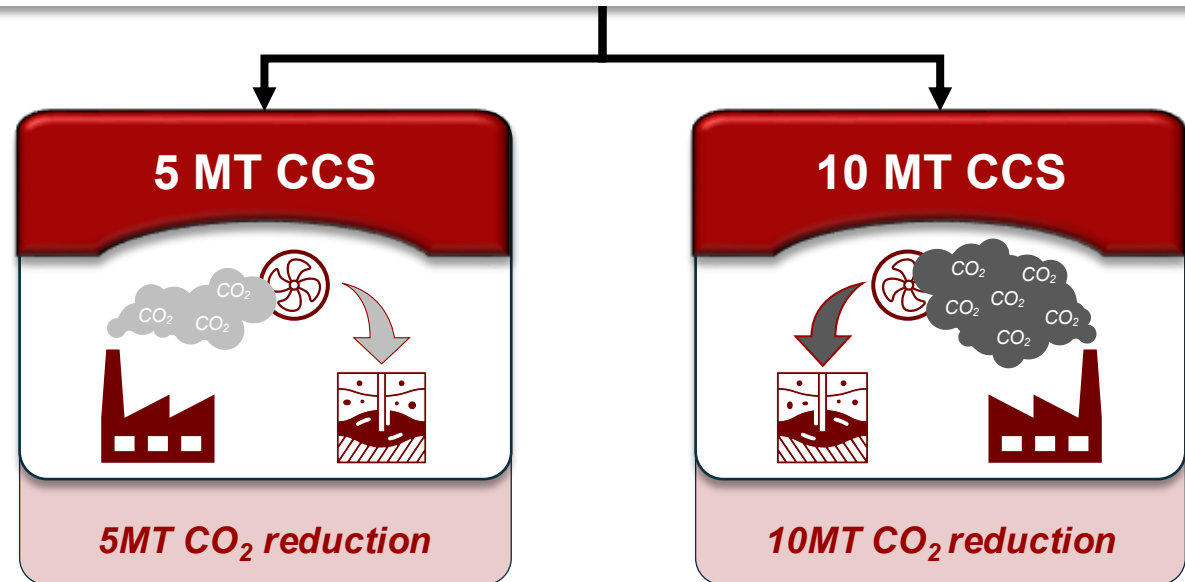
☐ Information Experiment

☒ Valuation Scenario(s)

☐ Valuation (WTP) Question

Valuation & Scope Scenario(s)

“(...) the Danish government wants to capture and transport CO₂ and store it in the North Sea (...)”



WTP for CCS Survey

Information Experiment

Valuation Scenario(s)

Valuation (WTP) Question

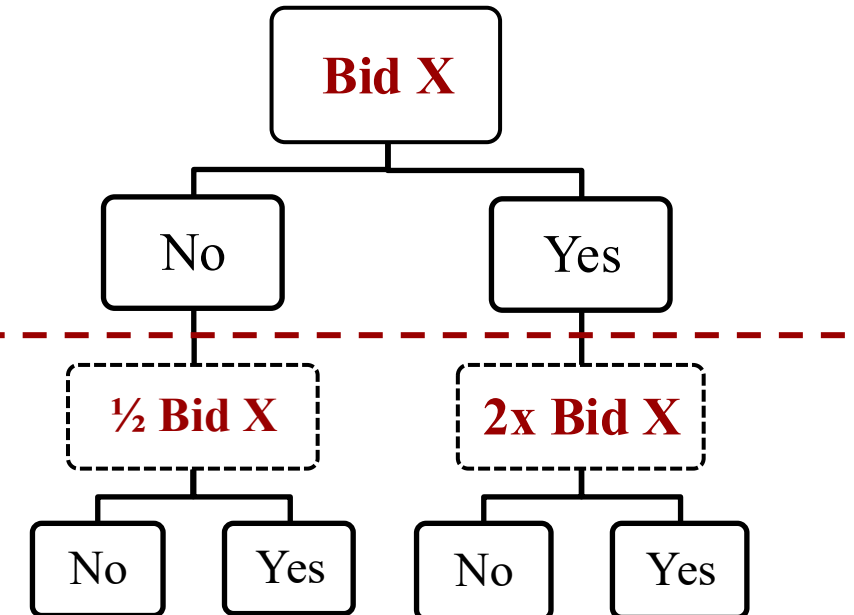
Valuation Question

"Would you be willing to pay (X kr.) to capture and store (5/10) million tons of CO₂ per year?"

- Contingent Valuation Method (CVM)
- Double-bounded dichotomous choice (DBDC):

1st Response Round

2nd Response Round



WTP for CCS Survey

Information Experiment

Valuation Scenario(s)

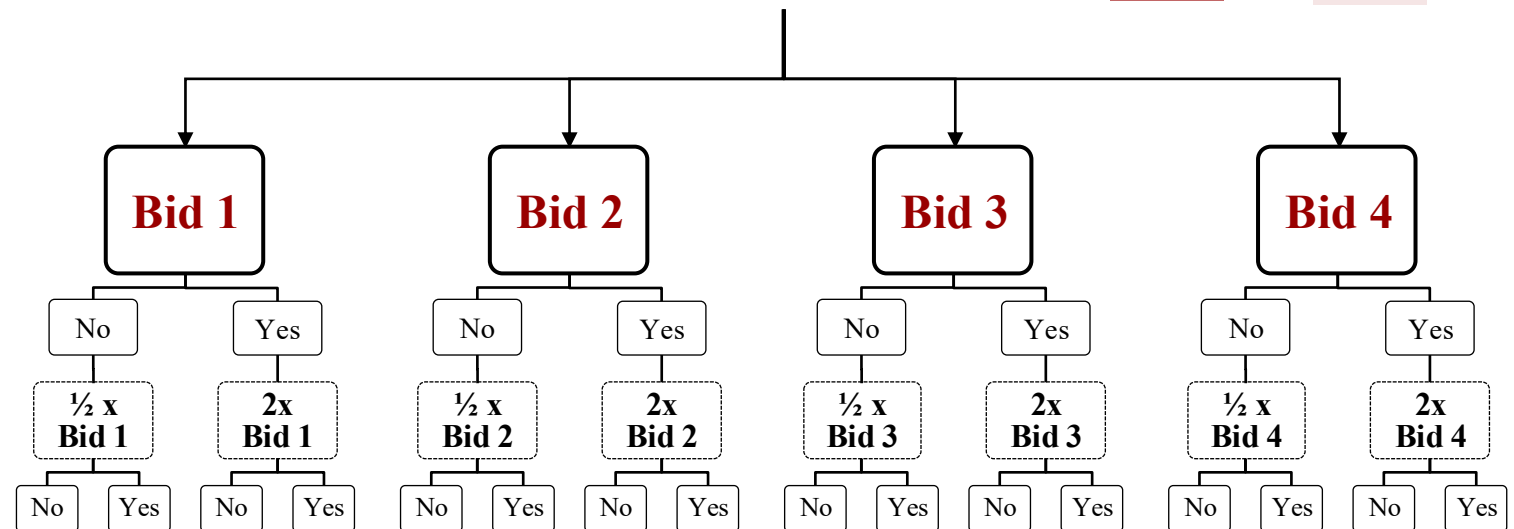
Valuation (WTP) Question

Valuation Question

"Would you be willing to pay (X kr.) to capture and store (5/10) million tons of CO₂ per year?"

Bid Vector:

- 4 starting bids, randomly assigned to respondents
- Bid levels conditional on CCS Scenario (**5MT** vs. **10MT**)



WTP for CCS Survey

Information Experiment

Valuation Scenario(s)

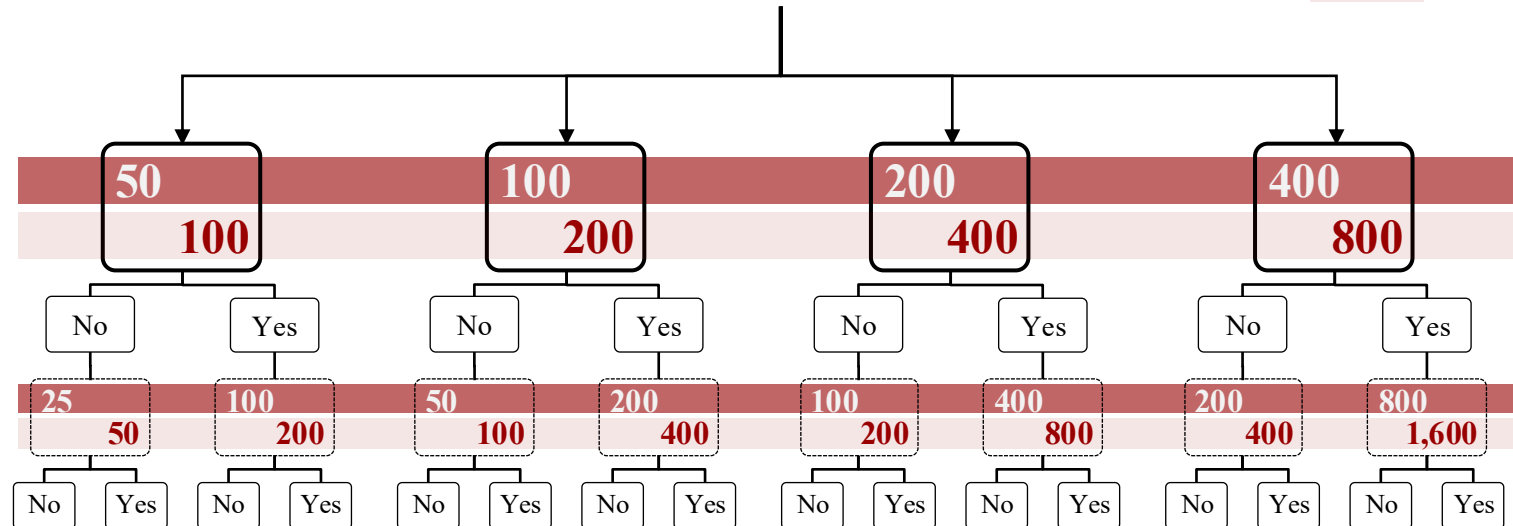
Valuation (WTP) Question

Valuation Question

"Would you be willing to pay (X kr.) to capture and store (5/10) million tons of CO₂ per year?"

Bid Vector:

- 4 starting bids, randomly assigned to respondents
- Bid levels conditional on CCS Scenario (**5MT** vs. **10MT**)



Information Effects on CCS Acceptance

Results: Information, order effects and CCS acceptance

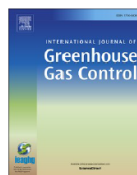
International Journal of Greenhouse Gas Control 150 (2026) 104566



Contents lists available at ScienceDirect

International Journal of Greenhouse Gas Control

journal homepage: www.elsevier.com/locate/ijggc



Can acceptance of location-specific carbon capture and storage be tipped?
The causal effects of information and question framing

Jacob Ladenburg ^{*}, Matteo Zuch ^{ib}, Jiwon Kim

Technical University of Denmark (DTU), Department of Technology, Management and Economics (DTU-MAN), Climate and Energy Policy Division, Produktionstorvet 424, 024, 2800 Kgs. Lyngby, Denmark

CCS Location	Offshore	Nearshore	Onshore Rural	Onshore Urban
Variables	Estimate	Estimate	Estimate	Estimate
INFO_A	0.057 (0.044)	0.117 * (0.045)	0.101 * (0.046)	0.068 (0.047)
INFO_DK	0.029 (0.045)	0.078 (0.046)	0.131 ** (0.047)	0.010 * (0.048)
CCS_FAM	0.097 * (0.039)	0.118 ** (0.041)	0.113 ** (0.042)	0.057 (0.042)
Q_ORDER	− 0.159 *** (0.037)	− 0.119 ** (0.038)	0.027 (0.039)	0.148 *** (0.039)
Constant	3.217 *** (0.124)	3.054 *** (0.124)	2.925 *** (0.126)	3.031 ** (0.130)
Socio Controls	Yes	Yes	Yes	Yes
Spatial Controls	Yes	Yes	Yes	Yes
N	3,877	3,878	3,879	3,877
R ²	0.024	0.015	0.023	0.023
LL(β)	− 6,034	− 6,145	− 6,185	− 6,250

Notes: *** p < 0.001, ** p < 0.01, * p < 0.05; Standard errors in parentheses

Information Effects on CCS Acceptance



Information provision

- Receiving additional information positively affects CCS acceptance (*relative to BI*)
- Information effects vary depending on the proposed CCS location:
 - a) Information about international CCS experience (**INFO_A**) increases acceptance of **nearshore** and **rural onshore** CCS
 - b) Information about Danish gas storage experience (**INFO_DK**) increases acceptance of onshore CCS (**rural** & **urban**)
 - c) No impact on **offshore CCS** acceptance

CCS Location	Offshore	Nearshore	Onshore Rural	Onshore Urban
Variables	Estimate	Estimate	Estimate	Estimate
INFO_A	0.057 (0.044)	0.117 * (0.045)	0.101 * (0.046)	0.068 (0.047)
INFO_DK	0.029 (0.045)	0.078 (0.046)	0.131 ** (0.047)	0.010 * (0.048)
CCS_FAM	0.097 * (0.039)	0.118 ** (0.041)	0.113 ** (0.042)	0.057 (0.042)
Q_ORDER	− 0.159 *** (0.037)	− 0.119 ** (0.038)	0.027 (0.039)	0.148 *** (0.039)
Constant	3.217 *** (0.124)	3.054 *** (0.124)	2.925 *** (0.126)	3.031 ** (0.130)
Socio Controls	Yes	Yes	Yes	Yes
Spatial Controls	Yes	Yes	Yes	Yes
N	3,877	3,878	3,879	3,877
R ²	0.024	0.015	0.023	0.023
LL(β)	− 6,034	− 6,145	− 6,185	− 6,250

Information Effects on CCS Acceptance



Prior CCS familiarity

- Prior CCS familiarity (**CCS_FAM**) increases CCS acceptance

CCS Location	Offshore	Nearshore	Onshore Rural	Onshore Urban
Variables	Estimate	Estimate	Estimate	Estimate
INFO_A	0.057 (0.044)	0.117 * (0.045)	0.101 * (0.046)	0.068 (0.047)
INFO_DK	0.029 (0.045)	0.078 (0.046)	0.131 ** (0.047)	0.010 * (0.048)
CCS_FAM	0.097 * (0.039)	0.118 ** (0.041)	0.113 ** (0.042)	0.057 (0.042)
Q_ORDER	− 0.159 *** (0.037)	− 0.119 ** (0.038)	0.027 (0.039)	0.148 *** (0.039)
Constant	3.217 *** (0.124)	3.054 *** (0.124)	2.925 *** (0.126)	3.031 ** (0.130)
Socio Controls	Yes	Yes	Yes	Yes
Spatial Controls	Yes	Yes	Yes	Yes
N	3,877	3,878	3,879	3,877
R ²	0.024	0.015	0.023	0.023
LL(β)	− 6,034	− 6,145	− 6,185	− 6,250

Notes: *** p < 0.001, ** p < 0.01, * p < 0.05; Standard errors in parentheses

Information Effects on CCS Acceptance



Prior CCS familiarity

- Prior CCS familiarity (**CCS_FAM**) increases CCS acceptance



Question order

- Strong question-ordering effects (**Q_ORDER**) on CCS acceptance
- Receiving onshore CCS options first decreases acceptance of **nearshore** and **offshore** CCS
 - But, increases **onshore urban** CCS acceptance
- **Priming effects on CCS acceptance across different placement options**

CCS Location	Offshore	Nearshore	Onshore Rural	Onshore Urban
Variables	Estimate	Estimate	Estimate	Estimate
INFO_A	0.057 (0.044)	0.117 * (0.045)	0.101 * (0.046)	0.068 (0.047)
INFO_DK	0.029 (0.045)	0.078 (0.046)	0.131 ** (0.047)	0.010 * (0.048)
CCS_FAM	0.097 * (0.039)	0.118 ** (0.041)	0.113 ** (0.042)	0.057 (0.042)
Q_ORDER	-0.159 *** (0.037)	-0.119 ** (0.038)	0.027 (0.039)	0.148 *** (0.039)
Constant	3.217 *** (0.124)	3.054 *** (0.124)	2.925 *** (0.126)	3.031 ** (0.130)
Socio Controls	Yes	Yes	Yes	Yes
Spatial Controls	Yes	Yes	Yes	Yes
N	3,877	3,878	3,879	3,877
R ²	0.024	0.015	0.023	0.023
LL(β)	- 6,034	- 6,145	- 6,185	- 6,250



Results: WTP for CCS

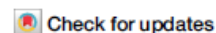
communications earth & environment

Article



<https://doi.org/10.1038/s43247-024-01900-y>

Public demand for carbon capture and storage varies with information, development magnitude and prior familiarity



Jiwon Kim¹ & Jacob Ladenburg²✉

Main regression results on WTP for CCS with 5 MT and 10 MT CO₂ reduction capacity; Split-sample analysis

WTP Scenario	5 MT CCS	10 MT CCS
Model	(I)	(I)
Variables	Estimate	Estimate
CCS_FAM	62.44^{***} (19.50)	76.07^{**} (30.58)
INFO_A	27.07 (21.37)	36.47 (35.92)
INFO_DK	49.68 ^{**} (22.35)	−40.77 (35.98)
Constant	246.1 ^{***} (75.77)	338.0 ^{**} (137.8)
Socio Controls	Yes	Yes
Spatial Controls	Yes	Yes
Fixed Effects	PZ	PZ
N	1,660	1,730
Wald Chi ²	172.8 ^{***}	206.3 ^{***}

Notes: *** p < 0.01, ** p < 0.05, * p < 0.1; Std. errors in brackets; PZ FEs

Insights from Preference Dimension



Prior CCS familiarity

- Familiarity with CCS (**CCS_FAM**) positively influences WTP for CCS

Main regression results on WTP for CCS with 5 MT and 10 MT CO₂ reduction capacity; Split-sample analysis

WTP Scenario	5 MT CCS	10 MT CCS
Model	(I)	(I)
Variables	Estimate	Estimate
CCS_FAM	62.44^{***} (19.50)	76.07^{**} (30.58)
INFO_A	27.07 (21.37)	36.47 (35.92)
INFO_DK	49.68 ^{**} (22.35)	−40.77 (35.98)
Constant	246.1 ^{***} (75.77)	338.0 ^{**} (137.8)
Socio Controls	Yes	Yes
Spatial Controls	Yes	Yes
Fixed Effects	PZ	PZ
N	1,660	1,730
Wald Chi ²	172.8 ^{***}	206.3 ^{***}

Insights from Preference Dimension



Prior CCS familiarity

- Familiarity with CCS (**CCS_FAM**) positively influences WTP for CCS



Information provision

- Info about Danish gas storage experience (**INFO_DK**) increases WTP for 5MT CCS
- The effect of information on WTP depends on the project's CO₂ storage capacity

Main regression results on WTP for CCS with 5 MT and 10 MT CO₂ reduction capacity; Split-sample analysis

WTP Scenario	5 MT CCS	10 MT CCS
Model	(I)	(I)
Variables	Estimate	Estimate
CCS_FAM	62.44 ^{***} (19.50)	76.07 ^{**} (30.58)
INFO_A	27.07 (21.37)	36.47 (35.92)
INFO_DK	49.68^{**} (22.35)	-40.77 (35.98)
Constant	246.1 ^{***} (75.77)	338.0 ^{**} (137.8)
Socio Controls	Yes	Yes
Spatial Controls	Yes	Yes
Fixed Effects	PZ	PZ
N	1,660	1,730
Wald Chi ²	172.8 ^{***}	206.3 ^{***}



“How and to what extent does information provision affect public responses to CCS?”

- 1) CCS-related information influences public responses to CCS development
 - Informing about existing projects and CCS experience can increase support
- 2) Information effects are highly context-sensitive as they depend on:
 - a) the information content provided,
 - b) the proposed location of CCS
 - c) the project's scope in terms of emission reduction potential
- 3) Experience-framed information has no impact on offshore **CCS acceptance** but significantly influences **preference-responses** for this siting option
 - Preferences for CCS might be more sensitive to external informational cues than more general acceptance statements

Conclusion

- Social acceptance of CCS is multifaceted, involving experience, knowledge, and intricate psychological factors.
- By understanding and addressing public concerns, fostering open dialogue, and involving communities, we can work towards a more sustainable and effective low-carbon future.
- Investigating heterogeneity in acceptance across socio-economic groups provides insights for tailored communication and engagement strategies.
- Acknowledging and addressing these determinants are essential for successful CCS implementation.

Exercises/group work

- Imagine that you are responsible for an acceptance study of the proposed energy island on Bornholm?
- Elaborate on the model by Huijts et al. (2012) and Zuch and Ladenburg (2023) in this relation.
- Which impacts/roles would the different factors, knowledge and information have in the case of an energy island at Bornholm?
- What would be important to inform people about in the case of an energy island at Bornholm if you wish to assess the assessment of the island?
- Who would you ask to answer your survey.
- Try to formulate an energy island acceptance question – including the information you wish the add prior to the question (framing)

Thank you for your attention

Contact information

Prof. Jacob Ladenburg

Email: jlad@dtu.dk

Phone: 42493610

References

- Ashworth, P., Bradbury, J., Wade, S., Ynke Feenstra, C.F.J., Greenberg, S., Hund, G., Mikunda, T., 2012. What's in store: Lessons from implementing CCS. *Int. J. Greenh. Gas Control* 9, 402–409. <https://doi.org/10.1016/j.ijggc.2012.04.012>
- Braun, C., Merk, C., Pönitzsch, G., Rehdanz, K., Schmidt, U., 2018. Public perception of climate engineering and carbon capture and storage in Germany: survey evidence. *Clim. Policy* 18, 471–84. <https://doi.org/10.1080/14693062.2017.1304888>
- Dillman, D.A., Smyth, J.D., Christian, L.M., 2014. *Internet, phone, mail, and mixed mode surveys: The tailored design method*, 4th ed., Internet, phone, mail, and mixed mode surveys: The tailored design method, 4th ed. John Wiley & Sons, Ltd.
- Gaede, J., Rowlands, I.H., 2018. Visualizing social acceptance research: A bibliometric review of the social acceptance literature for energy technology and fuels. *Energy Res. Soc. Sci.* <https://doi.org/10.1016/j.erss.2017.12.006>
- Greene, W.H., 2012. *Econometric analysis* 7th Ed, Prentice Hall. Prentice-Hall, New Jersey.
- Ha-Duong, M., Nadaï, A., Campos, A.S., 2009. A survey on the public perception of CCS in France. *Int. J. Greenh. Gas Control* 3, 633–640. <https://doi.org/10.1016/j.ijggc.2009.05.003>
- Huijts NMA, Molin EJE, Steg L. Psychological factors influencing sustainable energy technology acceptance: A review-based comprehensive framework. *Renewable and Sustainable Energy Reviews* 2012;16:525–31.
- Itaoka, K., Okuda, Y., Saito, A., Akai, M., 2009. Influential information and factors for social acceptance of CCS: The 2nd round survey of public opinion in Japan, in: *Energy Procedia*. pp. 4803–4810. <https://doi.org/10.1016/j.egypro.2009.02.307>
- Karimi, F., Toikka, A., 2014. The relation between cultural structures and risk perception: How does social acceptance of carbon capture and storage emerge?, in: *Energy Procedia*. pp. 7087–7095. <https://doi.org/10.1016/j.egypro.2014.11.743>
- Ladenburg, J., Kim, J., Zuch, M., Soytaş, U., 2023. Exploring the Acceptance of Landscape Specific Climate Change Mitigation Technologies – a Danish National Representative Study. Available at SSRN: <https://ssrn.com/abstract=4419472> or <http://dx.doi.org/10.2139/ssrn.4419472>
- Karytsas, S., Polyzou, O., Oikonomou, T.I., Karytsas, C., 2023. A transnational study on the determinants of social acceptance of carbon capture, transport, and storage (CCS). *IOP Conf. Ser. Earth Environ. Sci.* 1196, 1–12. <https://doi.org/10.1088/1755-1315/1196/1/012092>
- Matthew B.Miles, Saldana, J., Huberman, M.A., 2019. *Qualitative Data Analysis - A Methods Sourcebook*. Sage Publications Inc.
- Newcomer, K.E., Hatry, H.P., Wholey, J.S., 2015. *Handbook of practical program evaluation: Fourth edition*, *Handbook of Practical Program Evaluation: Fourth Edition*. <https://doi.org/10.1002/9781119171386>
- Oltra, C., Dütschke, E., Preuß, S., Gonçalves, L., Germán, S., 2022. What influences public attitudes and acceptance of CCUS technologies on the national and regional level? Results from a survey study in France and Spain. *SSRN Electron. J.* <https://doi.org/10.2139/ssrn.4272128>
- Patton, M.Q., 2014. *Qualitative Research & Evaluation Methods Integrating Theory and Practice*. SAGE Publications, Inc.
- Saunders, M., Lewis, P., Thornhill, a., 2009. *Research Methods for Business Students*, Business.
- Shackley, S., McLachlan, C., Gough, C., 2004. The public perception of carbon dioxide capture and storage in the UK: Results from focus groups and a survey. *Clim. Policy* 4, 377–398. <https://doi.org/10.1080/14693062.2004.9685532>
- von Rothkirch, J., Ejderyan, O., 2021. Anticipating the social fit of CCS projects by looking at place factors. *Int. J. Greenh. Gas Control* 110, 103399. <https://doi.org/10.1016/j.ijggc.2021.103399>
- Zuch, M. and J. Ladenburg (2023): Untangling the Threads: Finding Consistent Patterns of Information Effects on CCS Acceptance. Paper in review (minor revision) in *Energy Research & Social Science*. (Draft can be provided)